

Step 2: Create a Maven Project in VS Code

1. Press **Ctrl + Shift + P** to open the Command Palette.
2. Type and select **Java: Create Java Project....**
3. Choose **Maven** as the project type.
4. Choose **maven-archetype-quickstart**.
5. Select version **1.4**.
6. Set Group ID: **org.priyanshu**
7. Set Artifact ID: **WordCount**
8. Select a folder on your computer to store the project.
9. When prompted in the bottom terminal panel, press **Enter** to confirm project creation.
10. Click **Open** when VS Code asks to open the created project.

Configure **pom.xml**

Create Java Source Files

```
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount> mvn archetype:generate -DgroupId="org.granth" -DartifactId="wordcount" -DarchetypeArtifactId="maven-archetype-quickstart" -DarchetypeVersion="1.4" -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
```

```
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount> cd .\wordcount\
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.granth:wordcount >-----
```

```
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.granth:wordcount >-----
[INFO] Building wordcount 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
```

WC_Runner.java change

Set memory configurations programmatically to fit local YARN memory limit

```
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.granth:wordcount >-----
[INFO] Building wordcount 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ wordcount ---
```

```
[INFO] Finished at: 2026-09-23T02:00:51+05:30
[INFO]
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> hadoop jar target\wordcount-1.0-SNAPSHOT.jar org.granth.WC_Runner /Lab_3_BDA/input.txt /wordcount_output
The filename, directory name, or volume label syntax is incorrect.
2026-09-23 02:01:31 -0577 HADOOP client default configuration loaded successfully connecting to ResourceManager at localhost/127.0.0.1:8020
```

```
File Output Format Counters:  
    Bytes Written=37  
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> hadoop fs -cat /wordcount_output/*  
The filename, directory name, or volume label syntax is incorrect.  
hadoop 2  
hello 2  
mapreduce 1  
world 2  
PS D:\NITJ_Study\BDA_Lab\BDA_LAB_Github\Experiment-3\WordCount\wordcount> █
```